Smart Patch: A Non-Intrusive Wearable Technology for Healthcare Professionals

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Declaration

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Abstract

This design and build project delves into the application of non-intrusive wearable technologies for monitoring the health and well-being of healthcare professionals, particularly in high-stress environments like operating theatres. It responds to the imperative of safeguarding the health and safety of these professionals, recognizing their direct influence on patient outcomes. The project centers on integrating wearable technologies seamlessly into healthcare attire. These devices, interconnected via the Internet of Things (IoT), offer real-time monitoring of vital signs, facilitating prompt interventions as needed. Emphasizing a proof-of-concept approach, the project will tackle challenges of ensuring device accuracy, reliability, and maintaining non-intrusiveness while prioritizing healthcare professionals' well-being. Significantly, it aims to develop a centralized solution for healthcare institutions, employing Smart Patch—an IoT device—to provide continuous real-time monitoring of vital signs for all healthcare professionals. This centralized system will display comprehensive data on a dashboard, enhancing situational awareness and enabling proactive healthcare interventions. This research aims not only to contribute to the expanding realm of wearable technologies in healthcare but also to underscore their potential in elevating patient care by ensuring the sustained health and well-being of healthcare professionals.

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Chapter 1

Introduction

1.1 Motivation

Healthcare professionals (HCPs) are exposed to various sources of stress and pressure in their daily work, especially those who work in high-risk environments such as operation theatres. These stressors can negatively affect their physical and mental health and well-being, as well as their performance and quality of care. Extensive research has been conducted on this topic indicating that HCPs are liable to experience a range of mental health issues, including depression [1], anxiety [2], burnout [3], and stress [4]. A study which surveyed 3,700 public sector workers in the United Kingdom found that staff working for the National Health Service were the most stressed, with 61% reporting feeling stress all or most of the time and as of 2015 59% stating that stress has increased from previous years [5].

The well-being concerns for HCPs elevated when COVID-19 hit the world in 2019. A wealth of research was done around the world talking about anxiety, stress, depression and the psychological well-being of HCPs [6], [7]. This pressured healthcare institutions to work towards the health and well-being of their staff. More and more healthcare institutions worldwide are considering strategies for promoting a safety culture and staff well-being. This project aims to create a tool/device which helps these institutions get real-time insights into how to support HCPs in stressful environments best.

1.2 Aims and Objectives

The primary aim of this project is to delve into the potential of non-intrusive wearable technologies for real-time health monitoring of healthcare professionals, particularly those working in high-stress environments such as operating theatres, emergency rooms, and intensive care units. The motivation behind this aim is the critical need to ensure the health and well-being of these professionals, as their physical and mental state can directly impact the quality of care they provide to patients.

The secondary aim of this project (which could also be labelled as future work) is to open doors to further research opportunities that will elevate our current understanding of stressors in high-stress healthcare environments, and explore potential interventions to mitigate their impact. To reach that stage, a lot of potential work can be done following the creation of the device:

1. Conduct a feasibility study to evaluate the usability and acceptability of the prototype device among a sample of healthcare professionals working in high-stress environments, using surveys and interviews.
2. To analyse the data collected by the prototype device and assess the impact of the device on the health and well-being of healthcare professionals. This could involve statistical analysis of health indicators and correlation studies to understand the relationship between health indicators and stress and fatigue levels.
3. Based on the findings from the feasibility study and data analysis, provide recommendations and guidelines for the integration of non-intrusive wearable technologies into the existing attire of healthcare professionals and the improvement of their health and well-being. This could also include suggestions for improving HCPs' health and well-being based on the data analysis's insights.

If the timeframe of the project allows, the project will further dig into the above-mentioned works.

1.3 Proposed Solution: Smart Patch

This proposal outlines a solution to address the well-being concerns of HCPs by developing non-intrusive wearable technology for real-time monitoring. The proposed solution involves the development of a smart patch that can unobtrusively monitor key physiological indicators of well-being, i.e., heart rate variability and body temperature. The data collected from the patch will be transmitted wirelessly to a secure cloud platform, where it will be analyzed to provide personalized insights into the wearer's overall health status. These insights will be neatly displayed in a dashboard controller on the user's computer.

This project will include a web dashboard that interfaces with the device and allow a user to view relevant data wirelessly on their computers. With the information available at a glance, the hospital staff can take adequate measures to help the HCPs. More details of the requirements and implementation will be discussed in the subsequent chapters.

1.4 Overview of the Report

The introductory chapter establishes the foundation of this project, elucidating the motivation behind the study. It emphasizes the increasing awareness of healthcare professionals' health and well-being.

Chapter 2, delves into a literature survey, discussing the current strategies used by healthcare institutions and their inefficiencies. This chapter also discusses how the Internet of Things (IoT) came into the scenario and helped in digitalising healthcare but still has no solution that could help HCPs.

Chapter 3, navigates essential aspects of the project - essentially what will be developed. It provides an overview of how the system will work, what features will be implemented and success metrics. This chapter concludes with the risks identified.

Chapter 4, establishes the technical aspects of the project. Provides insights into what technologies will be needed for the success of the project. This chapter also includes a comparative study between various technologies for each component of the project and choosing the most suitable one for the project.

Chapter 5, provides an overview of the project's advancements, detailing key components that demonstrate the progress made.

Chapter 6, functions as a conclusion, summarizing significant findings and delineating future steps. In essence, this document combines elements of technology and health sciences, aligning with the interdisciplinary nature of health and well-being research.

Chapter 2

Literature Survey

This chapter serves as comprehensive exploration into existing strategies and methodologies within healthcare institutions aimed at supporting the well-being of healthcare professionals (HCPs). This chapter investigates current approaches and critically evaluates the limitations of these strategies in providing flexible and immediate support to HCPs. Additionally, it introduces the pivotal role of IoT and wearable technologies in revolutionizing healthcare monitoring. This chapter concludes by covering the limitations of these wearable technologies within delicate healthcare setting.

2.1 Current Strategies of Healthcare Institutions

The acknowledgement of these issues has prompted healthcare institutions to adopt various strategies to support the well-being of their staff.

Integrated Care Systems (ICSs): One notable approach is the implementation of Integrated Care Systems (ICSs). In the UK, the National Quality Board has introduced ICSs to enhance the quality of healthcare services. These systems aim to integrate planning and service delivery across different healthcare domains, including primary and specialist care, physical and mental health, and social care. By prioritizing self-care and prevention, ICSs aim to empower individuals to lead healthier and more independent lives [8].

Understanding Evolving Needs: In response to the evolving needs of HCPs, especially in the context of the COVID-19 pandemic, healthcare institutions have been exploring new ways to engage and support their staff. A survey of 720 healthcare providers conducted by Accenture in 2021 shed light on the changing expectations of HCPs. The findings emphasize that pharmaceutical companies need to better understand and meet these expectations to support healthcare professionals effectively in these challenging times [9].

Workplace Interventions: A systematic review published in the BMJ Open highlighted the impact of workplace interventions designed to improve well-being and reduce burnout among healthcare professionals. The interventions were primarily focused on managing stress in individuals, with mindfulness-based practices being adopted in 20 studies. Other interventions promoted a positive mindset (gratitude journaling, choirs, coaching), while organisational interventions centred on workload reduction, job crafting, and peer networks. [10]

Questionnaires: One step was to ask HCPs to fill out either self-administered or research-administered questionnaires to assess their health and well-being.

2.2 Inefficiency in Current Methodology

However, although the strategies might be well-researched and thorough they do have their limitations and don't provide flexibility in their strategy.

ICS: While ICSs aim to integrate planning and service delivery across different healthcare domains, they can sometimes lead to fragmented care from services that are not effectively coordinated around the needs of HCPs. This can negatively impact their experiences, lead to poorer outcomes, and create duplication and inefficiency [11]. Additionally, significant financial and resource inefficiencies had been recognized in the former Clinical Commissioning Groups (CCGs) before the implementation of ICSs. [12]

Understanding Evolving Needs: When understanding the evolving needs, While this strategy aims to adapt to the changing needs of healthcare professionals, it can sometimes be challenging to accurately identify and address these needs. The effectiveness of the strategy can be influenced by various factors, including the specific needs and circumstances of the HCPs and their healthcare settings. [13], [14]

Workplace Interventions: While workplace interventions can be effective in improving well-being and reducing burnout among healthcare professionals, the effectiveness of these interventions can vary depending on the specific intervention and the context in which it is implemented. Additionally, there can be challenges in ensuring that the interventions are appropriately tailored to the needs of the healthcare professionals and are implemented effectively. [15]

Questionnaire: They often fail to capture real-time changes in HCPs' health status and may yield misleading responses due to HCPs' tendency to prioritize patient care over their health because they often conceptualise their feelings as "in balance". For instance, respondents may provide incorrect feedback, and the response options may be limited. Additionally, questionnaires may lack flexibility, and the effectiveness of this strategy can be influenced by various factors, including the design of the questionnaire and the context in which it is administered.

These shortcomings underscore the need for more immediate and comprehensive methods of supporting HCPs. One modern way of solving this is introducing wearable technologies - an IoT device in healthcare institutions.

2.3 Internet of Things (IoT)

The Internet of Things (IoT) describes the network of physical objects—"things"—that are embedded with sensors, software, and other technologies to connect and exchange data with other devices and systems over the Internet. These devices range from ordinary household objects to sophisticated industrial tools. With more than 7 billion connected IoT devices today, experts are expecting this number to grow to 22 billion by 2025. [16]

While IoT has been in the tech industry for a a long time, only recent advances in several different technologies have made it practical and therefore popular.

IoT devices generate vast amounts of data - offering valuable insights for businesses and individuals. With advances in machine learning and analytics, coupled with IoT data, businesses can gather insights faster and more easily. The emergence of these allied technologies continues to push the boundaries of IoT and the data produced by IoT also feeds these technologies. [16]

2.4 Wearable Technology

Wearables are electronic devices (IoT devices) that can be worn or carried on the body. These devices can be separated into categories such as smartwatches, wristbands, and hearables. The personal data of the user can be monitored and measured through powerful microchips and smart sensors that are embedded within the wearable device. Connecting to additional devices using Bluetooth, Wi-Fi or a cellular network can further enhance the user experience [17].

Wearable technology has taken the world by storm, seamlessly integrating into our daily lives. These devices offer a wide range of functionalities, from tracking our physical activity and health metrics to providing us with real-time information and entertainment. As per Statista, 2023, smartwatches currently dominate the wearables market and have held a share of around 45 per cent since 2018. Most fitness trackers are built into smartwatches and can track steps, distance, heart rate, calories burnt, and other fitness metrics.

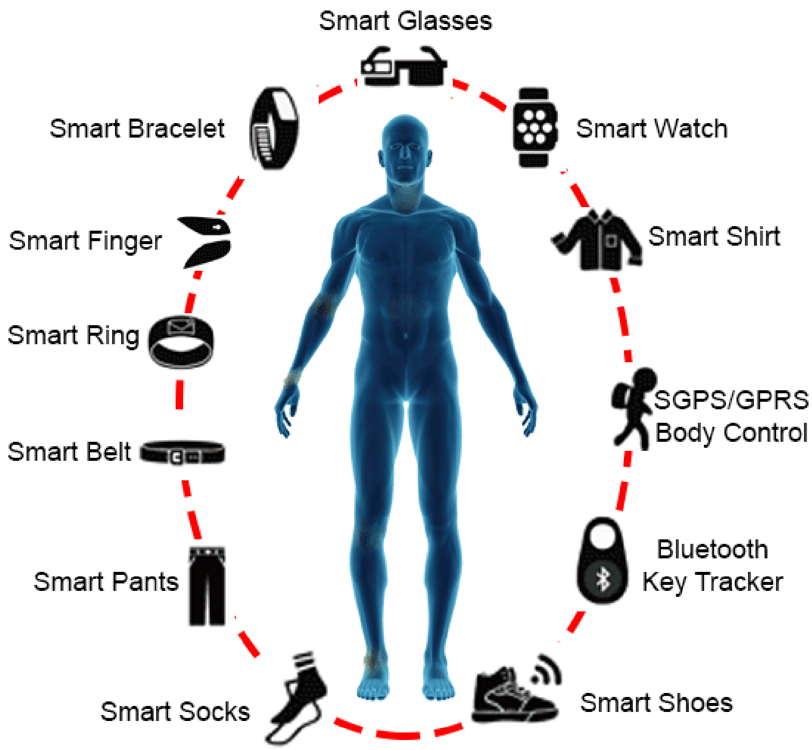


Figure 2.1: Wearable Technologies [18]

2.5 Limitations of Wearable Technology

Even though today's smart watches are state-of-the-art technology, HCPs are not allowed to wear them in delicate environments such as operation theatres or Intensive Care Units (ICUs). As per Uniforms and Workwear: Guidance for NHS Employers - NHS England, HCPs wearing watches at work is considered poor practice because they can harbour microorganisms making effective hand hygiene more difficult [19]. But it is not just that, wristwatches can potentially interfere with medical procedures. For instance, they could get in the way during surgery or delicate procedures.

ECG Patches were introduced in the healthcare market. ECG patches are adhesive sensors that are applied to the skin to monitor the electrical activity of the heart. They are typically used to record an electrocardiogram (ECG), which traces the heart's electrical activity. ECG patches can be used to monitor heart rate, rhythm, and variability. They are used mostly to monitor heart health in people with heart conditions and can provide real-time feedback. But these only limit to heart and related vitals which makes it unfit to calculate fatigue levels and therefore can't be used to solve the problem. In addition to that, there is so centralised system which can track these levels of all the personnel of an institution.

Chapter 3

Requirements and Analysis

In response to the imperative need for seamless healthcare monitoring, this project introduces the centralised system of Smart Patch and the dashboard for the institutions - an IoT innovation meticulously crafted to address the critical gaps in real-time healthcare professional (HCP) vitals monitoring. The chapter covers on the required features of the system and an evaluation system for the system is introduced.

3.1 Smart Patch

The Smart Patch is a non-intrusive wearable technology which will be designed to monitor the health and well-being of healthcare professionals in high-stress environments. This innovative IoT device aims to address the limitations of existing wearable technologies, particularly the restrictions on wearing smartwatches in delicate healthcare settings.

To develop this solution, it is crucial to identify and measure key parameters that reflect the health and well-being of HCPs. There are two main indicators which plays pivotal roles in the physiological and psychological aspects of an individual's health: stress level and fatigue level (further explained in **Appendix A**).

To effectively monitor stress and fatigue levels in real-time, the Smart Patch will utilize two key parameters:

Heart Rate Variability (HRV):

- HRV is a crucial indicator of the autonomic nervous system's activity and the body's ability to respond to stress.
- By analyzing the variation in time intervals between successive heartbeats, the smart patch can assess the balance between the sympathetic and parasympathetic nervous systems.

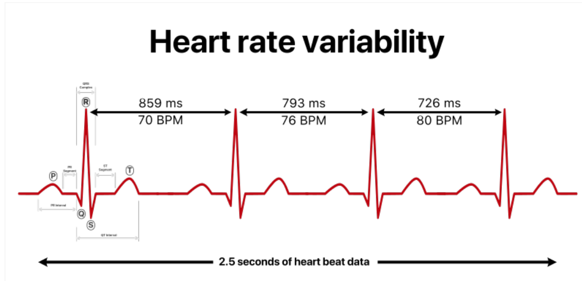


Figure 3.1: Heart Rate Variability [20]

- High HRV is associated with better adaptability to stress, while reduced HRV may indicate increased stress or fatigue.
- Real-time monitoring of HRV enables healthcare professionals to receive instant insights into the wearer's autonomic nervous system function.

Body Temperature:

- Monitoring body temperature in real-time provides valuable information about the wearer's overall health and potential signs of illness or stress. Elevated body temperature can be an early indicator of infections, inflammation, or increased physiological stress.
- In a healthcare setting, where infections can spread rapidly, early detection of abnormal body temperature can be crucial for preventing the spread of illnesses and ensuring the well-being of healthcare professionals.
- The smart patch will continuously track body temperature, allowing for prompt intervention and support in case of deviations from the normal range.

While stress and fatigue are complex and multifaceted, HRV and body temperature serve as surrogate markers that can be effectively measured in real-time. The integration of these parameters into the Smart Patch enables continuous monitoring, offering valuable insights into HCPs' health and well-being. By focusing on these indicators, the Smart Patch aims to provide actionable data that empowers healthcare institutions to proactively address stress and fatigue, ultimately enhancing the overall quality of care in high-stress environments.

3.2 Dashboard

To facilitate real-time monitoring, a web dashboard will be made where the institutions can track the vitals of all the HCPs working at that time using the data being collected by smart patch in real-time. It will present options to get alerts when an HCP's vital signs go beyond the threshold. Overall, the dashboard serves as a central hub for visualizing and interpreting the data collected by the smart patch, facilitating informed decision-making and proactive measures to support the health and well-being of HCPs in high-stress environments.

The Agile methodology of the Software Development Lifecycle (SDLC) (refer to **Appendix B**) will be employed for creating both the Smart Patch and the dashboard.

3.3 Features of the System

Both the Smart Patch and the Dashboard boasts a set of functional and non-functional features meticulously designed to ensure seamless real-time monitoring of healthcare professionals (HCPs). Both the features are categorised into two: functional and non-functional and further prioritsed using MoSCoW prioritisation framework into: Must-Have, Should-Have, Could-Have and Won't-Have. To see the entire list of features check **Appendix C**.

3.4 Evaluation

To ensure the robustness, reliability and effectiveness of the Smart Patch and the dashboard, a comprehensive testing and quality assurance process will be implemented throughout the implementation and testing stage of the project. Unit tests will be conducted to validate the correctness of individual components and the functionality of the Smart Patch and dashboard. To verify the seamless integration of components and modules within the Smart Patch and the dashboard, integration testing will be conducted. Finally, to assess the overall functionality and the performance of the integrated Smart Patch and the dashboard, system testing will be done.

To evaluate the usability of the Smart Patch and the dashboard, two interviews will be conducted later in the project: -

1. Dr. Naresh Chandra N Mehta MS, FMAS, FIMSA, FIAGES, FAIS - General and Laparoscopic Surgeon based in Surat, India. The interview will be conducted to gain insights into healthcare professionals' daily routines, challenges, and expectations, allowing for a user-centric design approach for the Smart Patch. In addition to this, to understand how Smart Patch fits into real-world scenarios. This insight is essential for refining the device's design to ensure its practicality and efficiency.
2. Mr. Vinod Choraria - Owner and Vice-Chairman of Nirmal Hospital Pvt. Ltd. based in Surat, India. He has experience working in hospital operations for more than 30 years. The interview will be conducted to gain insights into how the Smart Patch can seamlessly integrate into existing hospital operations and ensure minimal disruption to daily operations. The interview will also discuss the potential of adopting smart patches and the dashboard as a possible solution.

Chapter 4

Technology Stack

This chapter will cover on technology stack needed to build the smart patch and the dashboard. It will provide an overview of what might be needed, possible technologies to choose from and choosing the most suitable ones for the development of the system.

4.1 Hardware Components

4.1.1 Computing Units - Microcontrollers (MCU)

Given the proposed solution involves the integration of an embedded device, the computational units are typically configured in a compact form characterized by low power consumption and comparatively lower performance. This constraint directs attention to a specific category of single-board computing devices, wherein the board would contain essential components such as processor, memory and input/output interfaces, for running specific programs autonomously, obviating the necessity for supplementary expansion boards to fulfil these functions.

A microcontroller (MCU) is a compact, self-contained computer integrated on a single chip, designed for embedded applications with lightweight processing needs [21]. MCUs are known for their small size, low power consumption, and cost-effectiveness. They are widely utilized in various devices such as consumer electronics, automotive systems, industrial automation, medical devices, and the Internet of Things (IoT). MCUs require a hands-on approach, involving configuration for both hardware (attachment of components, sensors, wires, etc.) and software (burning firmware, incorporating libraries, installing drivers, etc.). MCUs lack a conventional operating system and utilize firmware, a special code burned onto the device to execute specific functions and ensure proper hardware functionality. While they cannot serve as personal computers, some advanced MCUs offer features like networking, making them appealing for projects with specific requirements.

There are multiple MCUs available in the market but the most popular ones are Raspberry Pi Pico and ESP32-WROOM-32S3-Mini-1. A comparative study between the two is provided

(a) Raspberry Pi Pico [22]

(b) ESP32 [23]

Figure 4.1: Comparison of Microcontrollers

in Table 4.1 to help in choosing the ideal MCU.

Feature	**Raspberry Pi Pico**	**ESP32-WROOM-32S3-Mini-1**
Manufacture	Raspberry Pi Foundation	Espressif Systems
Architecture	ARM Cortex-M0+	Xtensa LX6
Processor Clock Speed	Up to 133 MHz	Up to 240 MHz
Wireless Connectivity	None	Wi-Fi, Bluetooth (BLE)
Memory	2MB Flash (No built-in RAM)	2MB Flash
GPIO Pins	26	Approximately 38
Development Environment	MicroPython, C/C++ with Pico SDK	Arduino IDE, PlatformIO, Espressif IDF

Table 4.1: Comparative study between Raspberry Pi Pico and ESP32-WROOM-32S3-Mini-1

Comparing the two, ESP32-WROOM-32S3-Mini-1 comes out as ideal for the Smart Patch because of its in-built wireless connectivity. It is important to have such connectivity to provide portability of the device so that wearer can wear it all day and still can go around the campus.

4.1.2 Sensors

MCUs typically come equipped with general-purpose I/O pins (GPIO), allowing users to connect various external peripherals to the computer. Examples of common input devices include sensor modules, microphones, and cameras, while output devices encompass LEDs, speakers, displays, and more. Users can program these individual pins to receive data.

Heart Rate Sensors

To calculate HRV, heart rate sensors have to be used and connected with the MCU. Two sensors which are used abundantly in the industry, especially healthcare are ECG Sensors and Pimoroni Pulse Sensor. A comparative study is provided in Table 4.2 to help in determining the ideal sensor.

One of the main features of the Smart Patch is non-intrusivity. While both the sensors are efficient and used widely, Pulse Sensor comes out as a perfect choice for Smart Patch because not only it is less intrusive but also it works well with ESP32 (Arduino).

Body Temperature Sensor

To calculate body temperature, thermometers are used normally in healthcare. But thermometers are not digital devices and therefore can't be connected to the smart patch. Other sensors which are widely used in relation to sensing temperature are thermistors and LM-35 sensors. Thermistors are more temperature-sensitive resistors and not sensors and require a readout circuit, which makes it harder to integrate them seamlessly into Smart Patch. On the other hand, the LM-35 Sensor is a precision IC temperature sensor. It provides a linear voltage vs temperature characteristic making it easier to figure out the temperature. It can operate over a temperature range of 55°C to 150°C and has an accuracy of ±1 degree Celsius. Hence, LM-35 stands out as a clear choice for this project.

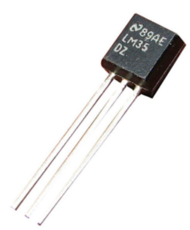


Figure 4.2: LM-35 Sensor [27]

4.1.3 Power Sources

An external power source is needed for the computing units to work. Since one of our major aims is to make the smart patch non-intrusive and therefore portable, it can't be connected to a power main source using USB ports. Batteries are a great alternative. They are portable as well as have low power consumption which allows for long battery life. Batteries can be categorized into two: primary cells and secondary cells. While primary cells are discarded once fully used, secondary cells are rechargeable. Since this project does consider sustainability, secondary cells become an obvious choice for the smart patch because the device will constantly be operating. While it's feasible to power Arduino/ESP boards with AA batteries, precautions must be taken to avoid underpowering or overpowering and damaging the board.

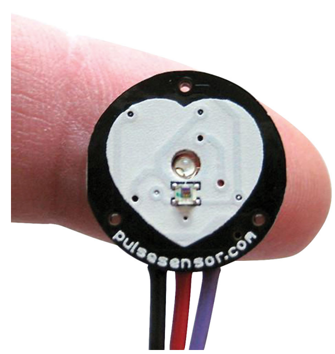
Features	**ECG Sensor**	**Pimoroni Pulse Sensor**
Working Principle	ECG Sensor measures the heart's electrical activity over time [24]. It detects the heartbeat rhythm regularly to keep a check on the health of your heart [25].	Pimoroni Pulse Sensor uses a technique called photoplethysmography (PPG) to detect the change in the colour of your skin with each beat of your heart when the sensor is pressed against your fingertip [26].
Use Cases	It can detect Arrhythmias like Atrial fibrillation and can also warn you about heart attack using strokes and movements around the heart.	It can be used by students, artists, athletes, makers, and game & mobile developers who want to easily incorporate live heart-rate data into their projects.
Compatibility	Works with most smartwatches.	Works with Arduino, micro:bit, littleBits, and most maker platforms.
Intrusivity	Non-invasive requires skin contact.	Less intrusive, fingertip contact.
Additional Features	Some devices can export an ECG graph of your heart rate, which can be a huge help when talking to your doctor.	The Pulse Sensor has 3 holes around the outside edge which make it easy to sew it into almost anything.
Image		

Table 4.2: Comparative Study between ECG Sensor and Pimoroni Pulse Sensor

Common rechargeable battery types include Lithium-Ion Polymer (LiPo) and Lithium-Ion. LiPo batteries, despite being slightly heavier due to construction, offer larger capacities. Despite structural differences, LiPo and Lithium-Ion batteries function similarly. Many boards feature a voltage regulator, which steps down input voltage to provide a stable DC output, releasing excess energy as heat.

The ESP32 can be connected via a Micro USB cable, with the board regulating the 5V USB down to 3.3V. Additionally, a 4.2/3.7V Lithium Polymer (LiPo/LiPoly) or Lithium-Ion (LiIon) battery can be connected via the JST jack.

Secondary batteries are more useful and can be utilised later in terms of scalability of the project. However, this project is more of a proof of concept and therefore it is easier to work with primary batteries. They are cheaper, easier to integrate and easily available.

4.1.4 Connectivity Technology

Connective Technology is nothing but a network of connected "smart" devices that communicate seamlessly over the internet. Wi-Fi, Bluetooth and LoRaWAN are the most used networks associated with IoT. Table 4.3 provides a comparative study between the three to help in choosing the most suitable technology for the project.

Features	**Wi-Fi**	**Bluetooth**	**LoRaWAN**
Scalability	Limited for massive IoT.	Mesh supports large networks.	Scales well for wide area.
Power Efficiency	High power consumption.	Low Energy, efficient.	Low power usage.
Range and Coverage	Short to medium range.	Moderate, indoor range.	Long-range, wide coverage.
Data Rate	High-speed data transfer	Moderate data rates	Low data rates
Cost and Infrastructure	Existing infrastructure may be costly.	Cost-effective, integrated widely.	Low infrastructure cost
Security	Robust encryption, authentication	Secure with enhancements	Implement AES encryption

Table 4.3: Comparative Study between Connective Technologies

From the comparison in Table 4.3 and Figure 4.3, LoRaWAN stands out as a clear and most efficient choice for Smart Patch. But the only disadvantage is that this network is not usually found in healthcare institutions compared to Wi-Fi which is available in nearly every healthcare institution. This advantage of Wi-Fi makes it easier to integrate Smart Patch with existing resources of institution thereby saving extra cost and other resources. Hence, Smart Patch will use Wi-Fi as its connective technology.

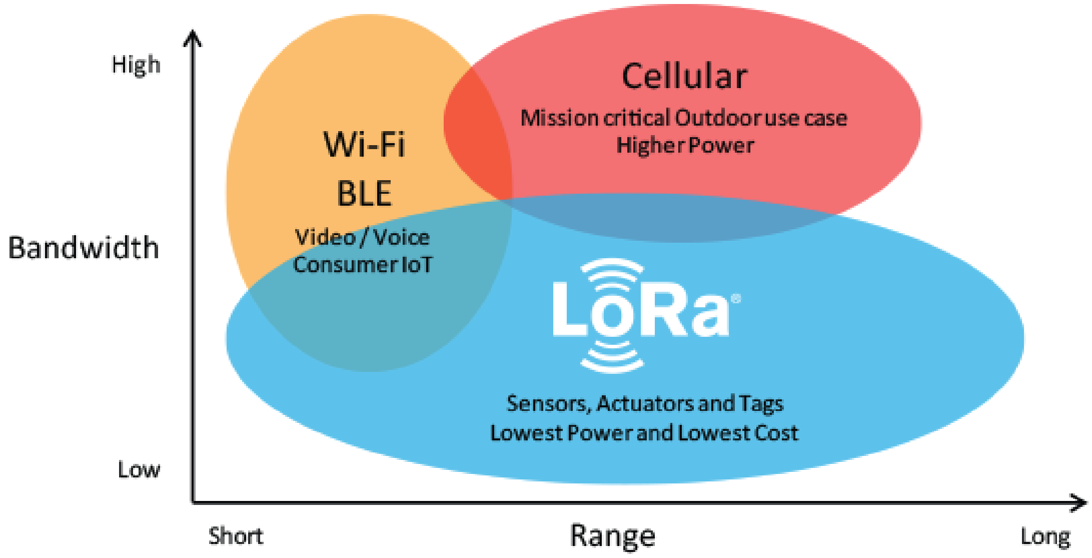


Figure 4.3: Connective Technologies compared against Bandwidth and Range [28]

4.2 Software and Application

4.2.1 Programming Environment

Arduino IDE, Espressif IDF (IoT Development Framework), and PlatformIO are three different programming environments used for developing software for microcontrollers and embedded systems. Each has its own strengths, weaknesses, and use cases.

	Arduino IDE	**Espressif IDF**	**PlatformIO**
Purpose	Beginner-friendly prototyping	Advanced ESP32/ESP8266 development	Cross-platform IoT development
Language Support	C/C++ with simplified syntax	Full C/C++ language support	Supports multiple languages
Hardware Compatibility	Extensive hardware compatibility	Specifically for ESP32/ESP8266	Broad hardware support
Integration	Integrated Development Environment	Integrated toolchain for ESP	Integrated platform and library manager

Table 4.4: Comparative Study between Programming Environments

After a thorough comparative study between the three (performed in Table 4.4), PlatformIO stands out as a clear choice for the project. It supports cross-platform development which is essential for the project as it also involves the development of dashboard. Additionally, PlatformIO supports multiple programming languages and broad hardware support which provides flexibility.

4.2.2 Programming Languages

ESP32 is compatible with 2 programming languages - Arduino C++ and MicroPython. Both of these languages are popular and used widely to program the firmware. The table 4.5 provides features of each of these languages.

Features	**Arduino C++**	**MicroPython**
Use	Firmware	Firmware
Programming Language	Low-level language with simplified syntax.	Python optimized for micro-controllers.
Abstraction level	Provides abstraction for hardware complexities.	Abstracts hardware with clean Python code.
Ease of Use	Beginner-friendly with a simplified structure.	Beginner-friendly, less complex than C/C++.
Memory Usage	Generally efficient memory usage.	May use more memory compared to C/C++.
Application Scope	Ideal for firmware development in embedded systems.	Suited for microcontroller programming.
Ecosystem	Rich ecosystem, extensive libraries.	Growing ecosystem for microcontrollers.
Concurrency Model	Sequential execution with setup() and loop().	Supports concurrency but less efficiently.
Frameworks	Arduino framework for embedded systems.	Limited frameworks for microcontrollers.
Resource Efficiency	Generally efficient in resource usage.	May consume more resources.

Table 4.5: Comparative Study between Arduino C++ and MicroPython

Both programming languages demonstrate efficiency in their respective ways. To determine the most suitable choice for the project, a decision matrix is employed (shown in Table 4.6). Each crucial feature is assessed for both languages, assigning a weighted score to each feature. Both scores and weights are provided in a range of 1 to 5.

Features	**Weight**	**Arduino C++**		**MicroPython**	
		Score	Weighted Score	Score	Weighted Score
Familiarity	3	2	6	4	12
Ease of Use	3	3	9	4	12
Memory	4	4	16	3	12
Ecosystem	3	4	12	3	9
Total			**43**		**45**

Table 4.6: Decision Matrix for Programming Languages

Following the completion of the decision matrix, it is determined that MicroPython emerges as the more fitting choice of programming language for the project.

Alternatively, a variety of IoT solutions are accessible in the market and offered at no cost. These options span from frameworks for creating dashboards to robust cloud platforms specifically designed for IoT applications. (see **Appendix E**)

For the Dashboard, JavaScript stands out as a widely favoured language in the realm of web development. However, relying solely on JavaScript may not harness the full potential needed for a robust and dynamic dashboard. This is where the MERN stack comes into play, offering a comprehensive solution for building feature-rich web applications. MERN, an acronym for MongoDB, ExpressJS, ReactJS, and NodeJS, represents a comprehensive set of technologies that synergize to streamline the development of feature-rich web applications. MongoDB offers a flexible NoSQL database, ExpressJS facilitates server-side code construction, ReactJS excels in creating dynamic user interfaces, and NodeJS provides the runtime environment for executing server-side JavaScript, all within a unified and efficient JavaScript ecosystem.

Within the project scope, a thorough examination of potential risks has been undertaken, delineating possible difficulties that might emerge throughout the project's duration. The risk register will be reviewed regularly throughout the project, allowing for the timely incorporation of newly recognised risks and fostering a proactive approach to risk management. The primary emphasis of this section is on identifying major potential risks and devising strategies to minimise their effects. **Appendix F** provides the risk register used for the project.

Chapter 5

Progress

This section provides a comprehensive overview of the progress made by the project and the ongoing activities of the project. Below are the listed:

- **Getting familiar with embedded systems:** Engaged in practical exercises from the Unphone: IoT book to build a foundational understanding of embedded systems.
- **Getting familiar with Platform IO and MicroPython:** Currently getting familiar with the development environment and the programming language for effectively programming the Smart Patch.
- **Training**
  - **iForge Training:** Completed iForge basic training and laser cut training. Will be finishing soldering training and 3D printing training by the end of December.
  - **Ethics Review Training:** Participated in ethics review training, ensuring that project activities aligned with ethical standards and guidelines.
- **Scheduling Interviews:** In the process of scheduling interviews and preparing a list of questions to be asked in the interview. Coordinating with the interviewers to have the interviews sometime in January.
- **Data Flow Diagram:** Developed a data flow diagram to visualize the movement and processing of information within the project, aiding in identifying key interactions and dependencies. Figure 5.1 represents how data will be transmitted from an HCP to Smart patch and Figure 5.2 represents how data collected by Smart patch will be transmitted to the dashboard in real-time.
- **Project Plan:** Established a comprehensive project plan using a Gantt Chart, providing a visual timeline of tasks and milestones to guide project execution and monitor progress. This project plan is laid out in Figure 5.3.

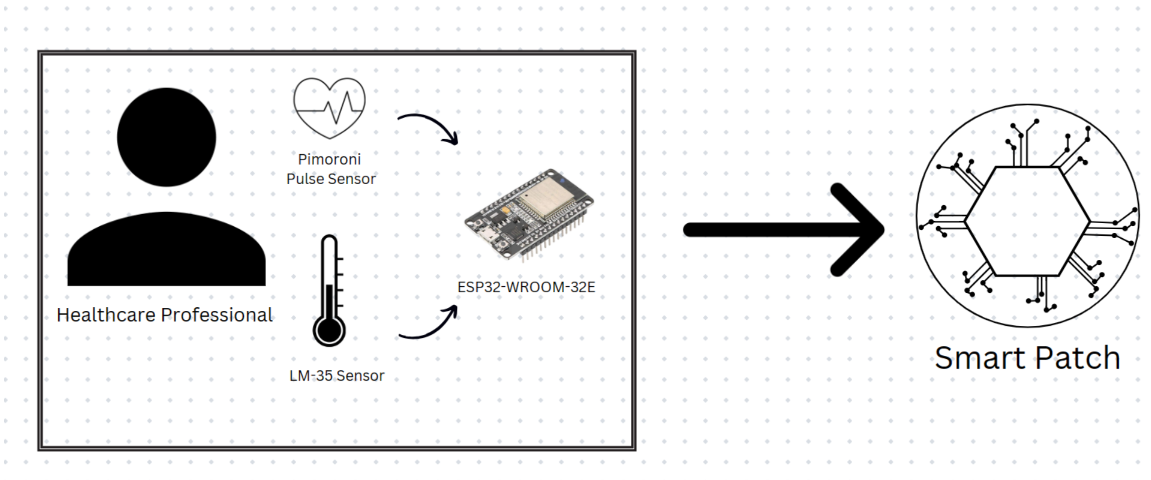


Figure 5.1: Data Flow Diagram - 1 (From HCP to Smart Patch)

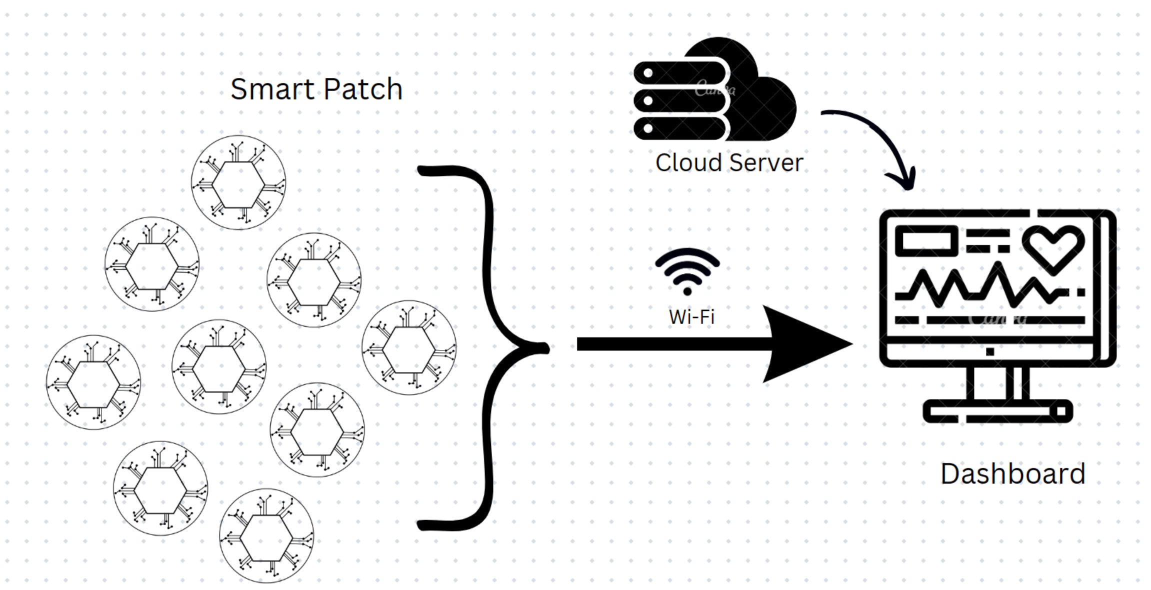


Figure 5.2: Data Flow Diagram - 2 (From Smart Patch to Dashboard)

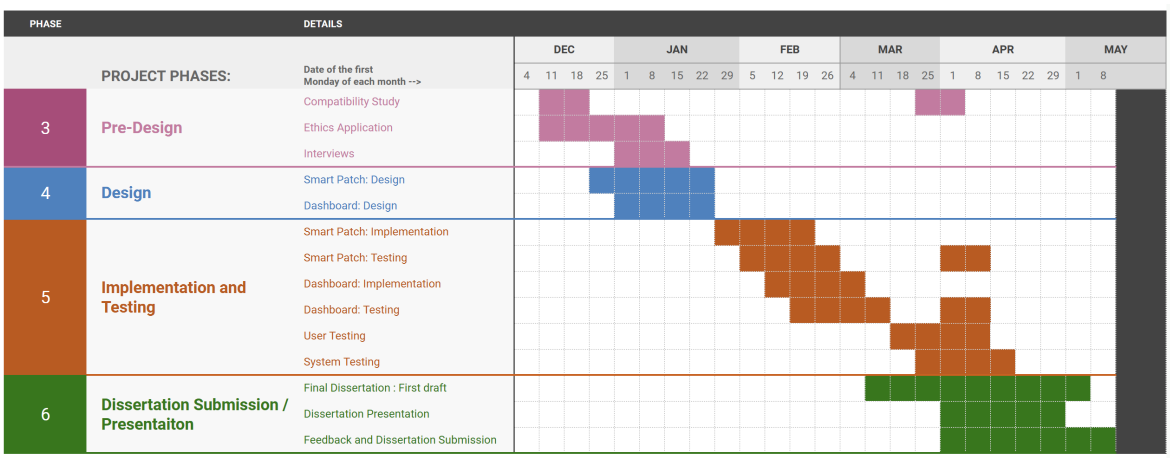


Figure 5.3: Project Plan

Chapter 6

Conclusion

With the imperative need for healthcare monitoring for healthcare professionals, this project aims to provide a solution where all the HCPs can be monitored non-intrusively and provide a centralized dashboard for the healthcare institutions to monitor the same. For this, the project will delve into two things:

- **Development of Smart Patch:** Smart Patch will be an IoT device built on ESP32, sending real-time heart beat rate and body temperature of HCP using Pimoroni Pulse Sensor and LM-35 Sensor, over Wi-Fi to the dashboard; programmed in MicroPython.

- **Development of Dashboard:** Dashboard will be a web interface developed in MERN stack technology providing real-time vitals information provided by the Smart Patch to healthcare institutions.

The project has surveyed the current methodologies, and technology which will be used to build the Smart Patch and the dashboard.

In the next phase of the project, the whole system will be designed, implemented and tested thoroughly and will be evaluated and delivered. Along with this, a few interviews will be conducted with healthcare professionals and healthcare institutions.

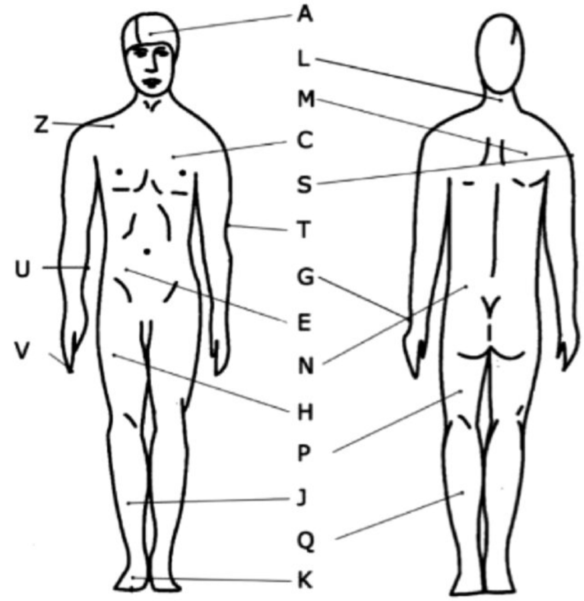


Figure 6.1: Enter Caption

Body Placement

Where exactly should this device be placed?

Why is it important?

Non-Intrusivity and Accuracy

Non-Intrusivity - doesn't bother them while doing an important work Places where it can be non-intrusive - back of neck/shoulders, behind ears, ankles

Places where we can measure body heart rate using pimoroni pulse sensor – 3d body image where all the places are marked

Places where we can measure body temperature using LM-35 sensor – 3d body image where all the places are marked

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Appendices

Appendix A

Stress and Fatigue Level

A.1 Stress Level:

Stress is a complex physiological and psychological response to external pressures or demands, commonly experienced in high-stress environments such as healthcare settings. Measuring stress levels involves assessing the body's reaction to stressors, which can manifest in various ways, including changes in heart rate, hormonal balance, and nervous system activity. The challenge lies in quantifying stress in real-time due to its multifaceted nature.

A.2 Fatigue Level:

Fatigue, often synonymous with tiredness, is a state of extreme tiredness or exhaustion, that impacts both physical and mental capacities. In healthcare professionals working in demanding environments, fatigue can compromise decision-making abilities, reaction times, and overall performance. Like stress, fatigue is a nuanced parameter that requires a comprehensive understanding of physiological and psychological factors.

Appendix B

Software Development Lifecycle (SDLC)

Agile Methodology is a popular approach in SDLC that emphasizes flexibility, collaboration and customer satisfaction [29] [30]. The development process is broken down into sprints (small, incremental builds or iterations). Such an iterative process allows developers to focus on the most important features at any given moment, rather than strictly following a pre-defined plan.

Agile Methodology is guided by its four core values [31]:

1. **Individuals and interactions** over processes and tools
2. **Working software** over comprehensive documentation
3. **Customer Collaboration** over contract negotiations
4. **Responding to change** over following a plan

The adoption of this methodology especially in IoT and software development has been widespread for several reasons.

- **IoT:** Agile methodologies support multiple protocols and networks utilised for building IoT solutions. It enables closer cooperation between IoT hardware and software development teams, fostering higher transparency [29]. This approach successfully addresses the complexity of connecting culture and technology in IoT implementation [33].
- **Software Development:** As per the 16th State of Agile Report by digital.ai, 78% of the respondents report that they have implemented or are planning to implement Agile at scale. Agile's emphasis on building working software quickly fostering cross-functional teams empowered to make decisions, and focusing on rapid iteration with continuous customer input are some of the reasons for its widespread adoption [34]. This approach not only follows better collaboration and incremental delivery but also provides early error detection and elimination of unnecessary work.

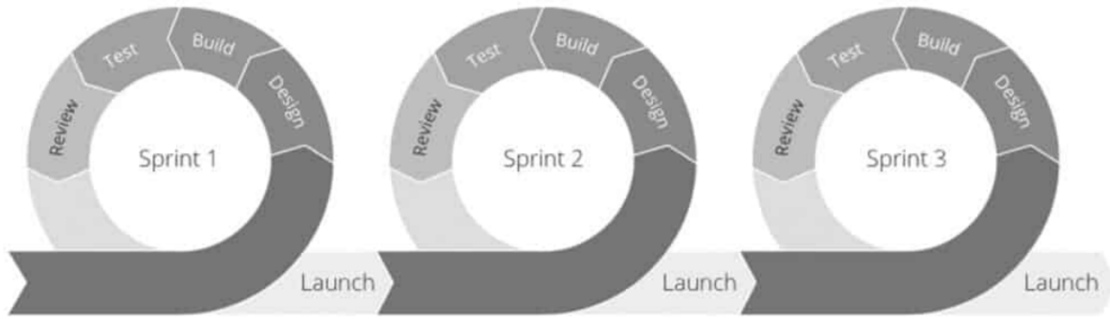

Figure B.1: Anatomy of Agile Approach [32]

Looking at the benefits of this approach, this project follows a similar approach.

Appendix C

Features of the System

- **Categorizing the Features**
  All the features of the system are categorized into functional and non-functional features.
  - **Functional Features**
    Functional features represent the capabilities and behaviours of a system that directly contribute to its primary purpose and functionality. These features are concerned with what the system does and how well it performs its intended tasks.
  - **Non-Functional Features**
    Non-functional features, on the other hand, are characteristics of a system that do not directly relate to its specific functionalities but rather focus on how well the system performs its functions.
- **MoSCoW Prioritisation FrameWork**
  After categorizing features into functional and non-functional, they are prioritized using the MoSCoW Prioritisation FrameWork. The key for this prioritisation is given in Table C.1.

Category	Description
Must-Have	This category consists of features that are "must" for the system. They represent non-negotiable needs for the system.
Should-Have	They are essential for the system, but they are not vital. If left out, the system still functions. However, the features may add significant value.
Could-Have	These features are not necessary to the core function of the system.
Won't-Have	This category contains features which will not be included in a specific release, time frame or prioritised in the future. Their absence is not detrimental to the functionality of the final product.

Table C.1: Key to features

C.1 Features of Smart Patch

C.1.1 Functional Features

Feature	**Description**	**MoSCoW Priority**
Real-time Monitoring	Continuously monitors vital signs (HRV and body temperature) in real-time, providing instant feedback on the wearer's physiological status.	Must-Have
Heart Beat Rate Monitoring	Monitor heart bear rate of the wearer in real-time.	Must-Have
Body Temperature Monitoring	Monitors body temperature of the wearer in real-time.	Must-Have
Wireless Connectivity	Equipped with wireless capabilities to securely transmit data to a cloud platform, enabling remote access for healthcare institutions.	Must-Have

Table C.2: Functional Features of Smart Patch

C.1.2 Non-Functional Features

Feature	**Description**	**MoSCoW Priority**
Non-Intrusiveness	Designed to be comfortable for prolonged wear, lightweight, unobtrusive, and adhering to safety and hygiene standards. Placed on the body where least intrusive for healthcare professionals.	Must-Have
Device Portability	Designed to be easily portable, allowing healthcare professionals to move freely and perform their duties without hindrance.	Must-Have
Regulatory Compliance	Adheres to relevant healthcare and IoT regulatory standards to ensure the device's safety, effectiveness, and compliance with legal requirements.	Must-Have
Long Battery Life	Designed with a long-lasting battery to minimise the need for frequent recharging, ensuring continuous monitoring during extended shifts.	Should-Have
Durability	Constructed with durable materials to withstand daily wear and tear in a healthcare environment, ensuring the device's longevity and reliability.	Should-Have
Compatibility	Ensures compatibility with various body types and sizes, accommodating diverse healthcare professionals with different preferences and needs.	Should-Have

Table C.3: Non-Functional Features of Smart Patch

C.2 Features of Dashboard

C.2.1 Functional Features

Feature	**Description**	**MoSCoW Priority**
Real-time Monitoring	Displays real-time heart rate and body temperature data of HCPs, allowing institutions to track vital signs continuously.	Must-Have
Alerts and Threshold	Enables institutions to set and receive alerts when an HCP's vital signs surpass predefined thresholds, facilitating timely intervention in case of abnormal patterns.	Must-have
Multiple HCP Tracking	Supports simultaneous monitoring of multiple HCPs, providing a comprehensive overview of the health status of all professionals working at a given time.	Should-Have
User Authentication	Implements secure user authentication to ensure authorized access, maintaining the confidentiality and integrity of healthcare data.	Could-Have
Customizable User Permissions	Provides customizable user permissions, allowing institutions to control access levels for different roles, ensuring data privacy and security compliance.	Could-Have

Table C.4: Functional Features of Dashboard

C.2.2 Non-Functional Features

Feature	**Description**	**MoSCoW Priority**
User-Friendly Interface	Designs an intuitive and user-friendly dashboard interface, enhancing the overall experience for healthcare professionals and administrators interacting with the monitoring system.	Must-Have
Performance Optimisation	Optimizes the dashboard's performance to handle real-time data efficiently, ensuring a smooth and responsive user experience even during periods of high data traffic.	Should-Have
Scalability	Designs the dashboard to be scalable, accommodating potential future expansions in the number of monitored healthcare professionals without compromising performance or user experience.	Should-Have
Security and Compliance	Ensure data security by implementing robust measures to protect sensitive healthcare information, complying with relevant regulations and industry standards.	Should-Have

Table C.5: Non-Functional Features of Dashboard

Appendix D

Sensors

To understand how sensors work, this section provides an overview of the colour coding of the sensors and the circuit diagram of each sensor used in the project.

D.1 Pulse Sensor Circuit

Figure D.2 shows the internal circuit diagram of a pulse sensor. It consists of an optical heartbeat sensor, an amplification circuit, and a noise cancellation circuit.

D.2 LM-35 Sensor Circuit

Figure D.3 shows the internal circuit diagram of a LM-35 sensor. In the figure, A1 shows the input where it senses the temperature and A2 provides the output voltage of the circuit. The circuit is made in such a way that it outputs a voltage proportional to the temperature it measures. This makes it easier to calculate the temperature of body eliminating electronic complexities.

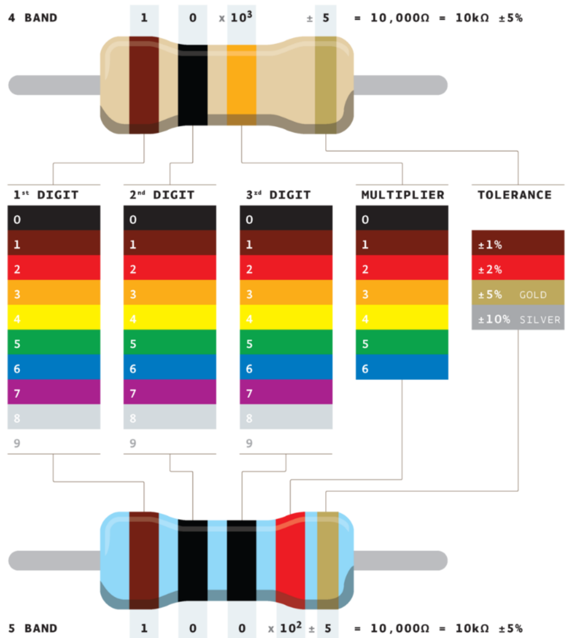

Figure D.1: Colour codes in sensors [35]

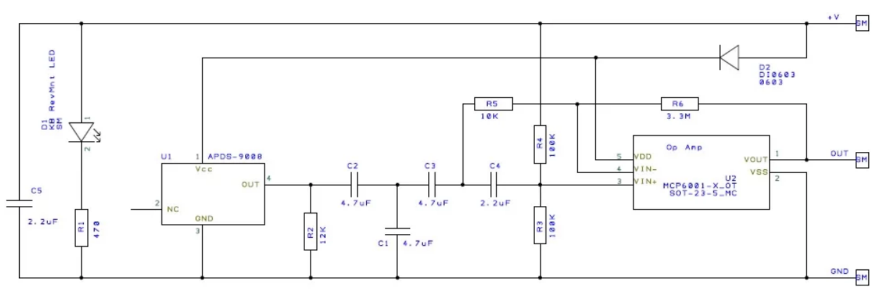

Figure D.2: Internal block diagram of an SEN-11574 pulse sensor [36]

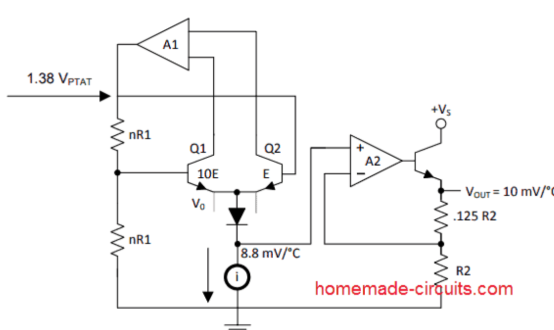

Figure D.3: Internal block diagram of IC LM-35 Sensor [37]

Appendix E

Similar Software Available for IoT Development

A number of softwares are available in the market for free of cost which supports IoT development. Below are some of the most commonly-used software for such development. If in the later phases of the project, it is find that it is efficient to use these instead of native development, they will be adopted.

1. **The Things Network (TTN)**
   Used: IoT cloud platform
   TTN is an open-source, decentralized infrastructure dedicated to IoT. TTN uses LoRaWAN as its secure messaging protocol. Devices that support LoRa can connect to gateways scattered all around the world that bridge them with TTN. It enables community users to create applications for data viewing and interact with the service by sending data from other IoT devices to other registered devices and gateways.

2. **ThingsBoard**
   Used: Dashboard tool
   ThingsBoard is an open-source IoT platform for data collection, processing, visualization, and device management. It enables device connectivity via industry standard IoT protocols - MQTT, CoAP and HTTP and supports both cloud and on-premises deployments. The benefits of using ThingsBoard are scalability, fault-tolerance and performance so that data is intact. [38]

3. **ThingSpeak**
   Used: Analysis and Visualization
   For even more advanced functionalities, there is ThingSpeak which is an IoT analytics platform service. This service allows a user to aggregate, visualize and even analyse live data streams without having to write any code. What makes ThingSpeak special is that it supports MATLAB analytics functions; users can write and run MATLAB

code for more advanced analyses. ThingSpeak can also integrate with TTN to retrieve data from connected devices. [39]

4. **Adafruit IO**
   Used: IoT cloud platform
   Adafruit IO is a cloud service. It can display data online from sensors in real-time, and connect to devices and control components. With a dashboard creator feature built-in, the software is capable of handling and visualizing multiple feeds of data. Adafruit IO can also be integrated with applications such as IFTTT and Zapier and other web services such as Twitter, RSS feeds etc. [40]

Appendix F

Risk Analysis

This chapter lists down all the potential risks that this project might face. The risk register provided in Figure F.2 is continuously updated throughout the project timeline as per the key to risk rating provided in Figure in F.1.

Key to Risk Rating:

Likelihood (L)	Description
5	Very likely/imminent - certain to happen.
4	Probable - a strong possibility of it happening
3	Possible - it may have happened before
2	Unlikely - could happen but unusual
1	Rare - highly unlikely to occur

Impact (I)	Description
5	Catastrophic:non-functional system
4	Major - significant changes to the system
3	Moderate - moderate changes to the system
2	Minor - minor changes to the system
1	Very minor - very minor changes to the system or requirements.

		Impact (I)				
		1	2	3	4	5
Likelihood (L)	5	5	10	15	20	25
	4	4	8	12	16	20
	3	3	6	9	12	15
	2	2	4	6	8	10
	1	1	2	3	4	5

Risk Rating (RR)	Action
High Risk	Stop the task/activity until controls can be put into place to reduce the risk to an acceptable level.
Medium Risk	Determine if further precautions are required to reduce risk to as low as is reasonably practicable.
Low Risk	No further action, keep under review.

Figure F.1: Key to Risk Rating

Risks	Consequences	Risk Rating			Control Measures	Residual Risk		
		L	I	RR		L	I	RR
Inadequate defines requirements	Can lead to missing core features and can result in a unsuccessful project	4	5	20	Ensure that the requirements are well-defined. Following an iterative development will allow to incorporate last minute changes	2	1	2
Lack of on-time feedback from the supervisor	Can potentially impact the quality of the project	3	4	12	Submitting drafts at least a week before the deadline. Setting up regular meetings This would give both parties enough time time to suggest/make changes	1	4	1
Extenuating circumstances	Can potentially lead to a delay	3	3	9	Communicate with you supervisor and take appropriate steps (extending the deadline)	1	3	3
Delay in shipment of sensors	Can potentially lead to a delay	4	4	16	Ordering sensors before actual implementation phase. Can contact another supplier for immediate delivery.	2	2	4
Issues with IDE	Can lead to a delay in the development stage	4	5	20	Checking version compatibilites. Using the most up-to-date versions - this would make sure that bug fixes are applied immediately.	3	2	6
Ethics Application being rejected	This may lead to extra work (making a second application) and can also lead to delay testing.	4	4	16	Starting the ethics application as soon as possible	2	2	4
Compatibility Issues between ESP32 and other sensors	This may take extra time to solve the issues or find another component to replace the faulty ones.	3	5	10	Compatibility study will be done before starting actual implementation. Will ask the supervisor to help me with it, refer to documentations to debug the issue. If not, will look out for other compatible components and order it ASAP.	3	2	6

Figure F.2: Risk Register

Appendix G

Abbreviations

AA - Double-A (battery size)
BLE - Bluetooth Low Energy
BMJ - British Medical Journal
CoAP - Constrained Application Protocol
GPIO - General Purpose Input/Output
HTTP - Hypertext Transfer Protocol
HCp - Healthcare Professionals
I/O or IO - Input/Output
IC - Integrated Circuit
IDE - Integrated Development Environment
IDF - IoT Development Framework
ICU - Intensive Care Unit
IoT - Internet of Things
IVD - In-Vitro Devices
LiIon - Lithium-Ion
LiPo - Lithium Polymer
LoRaWAN - Long Range Wide Area Network
MCU - Microcontroller Unit
MERN - MongoDB, ExpressJS, ReactJS, NodeJS
MQTT - Message Queuing Telemetry Transport
NoSQL - Not Only SQL (database)
OS - Operating System
PPG - Photoplethysmography
RAM - Random Access Memory
ROM - Read Only Memory
SDLC - Software Development Lifecycle
TTN - The Things Network
USB - Universal Serial Bus